Sensor Fusion-Based Tank Sloshing Compensation Using Adaptive Kalman Filtering

Samiksha Satre[1]

[1]Department of Electronics and Telecommunication Engineering,
N. K. Orchid College of Engineering & Technology, Solapur, India
[2]ISAE-SUPMÉCA (Institut Supérieur de Mécanique de Paris),
Saint-Ouen, France

Abstract

Accurate liquid level measurement in partially filled tanks becomes critically challenging under dynamic motion conditions due to sloshing effects. These disturbances introduce nonlinear noise, oscillatory artifacts, and phase delays in sensor readings, leading to unreliable measurements that compromise control system performance. This paper presents a comprehensive simulation-based study for sloshing compensation using multiple estimation techniques. Traditional filtering methods, including moving average filters, are systematically compared with adaptive Kalman filtering approaches. An adaptive Kalman filter (AKF) is proposed in which the measurement noise covariance is dynamically adjusted using inertial measurement unit (IMU)-based motion information. Rather than explicitly modeling complex free-surface fluid dynamics, the system adapts sensor trust based on instantaneous motion intensity, resulting in improved stability, accuracy, and real-time applicability. Simulation results across multiple motion profiles demonstrate that the adaptive Kalman approach achieves a root-mean-square error (RMSE) reduction of 78.2% compared to raw sensor data, providing a superior balance between noise rejection and transient responsiveness.

Keywords: sloshing compensation; adaptive Kalman filter; sensor fusion; inertial measurement unit; liquid level estimation; signal processing

1 Introduction

Liquid sloshing in partially filled tanks is a pervasive phenomenon encountered across diverse engineering domains, including automotive fuel systems [1], spacecraft propellant management [3], mobile robotics [13], and industrial chemical transport [2]. When subjected to external excitations such as acceleration, braking, cornering, or mechanical vibration, the free liquid surface undergoes complex oscillatory motion. These oscillations corrupt level-sensor readings, introducing measurement artifacts that propagate through downstream estimation and control algorithms.

Conventional approaches to sloshing compensation fall into two broad categories: (i) physics-based modeling, which relies on computational fluid dynamics (CFD) or equivalent mechanical models (e.g., pendulum or spring–mass–damper analogies) [1, 2], and (ii) signal-processing techniques that treat sloshing as a disturbance to be filtered from sensor data [8]. Physics-based models, while accurate, demand significant computational resources and precise knowledge of tank geometry and fluid properties, limiting their suitability for real-time embedded applications. Pure signal-processing methods, conversely, often sacrifice responsiveness for noise attenuation.

A third category, which forms the theoretical foundation of this work, involves state estimation techniques that account for biases and unknown inputs. Friedland [16] pioneered the treatment of bias in recursive filtering, later extending the framework to handle irreducible uncertainty [17] and developing the separate-bias estimation methodology [18]. These foundational contributions established that biases can be estimated separately from the main state vector, reducing computational burden while maintaining optimality.

This work bridges the gap between these paradigms by proposing an *adaptive Kalman fil-*

ter (AKF) that leverages inertial measurement unit (IMU) data to modulate sensor trust in real time. The key contributions of this paper are as follows:

1. A parameterized sloshing simulation model incorporating damped oscillatory dynamics coupled with realistic sensor noise profiles.
2. A systematic benchmarking framework comparing moving average and adaptive Kalman filters.
3. An IMU-driven adaptive measurement noise covariance scheme that yields superior RMSE performance without explicit fluid-dynamic modeling.
4. A detailed sensitivity analysis of filter parameters and motion intensity on estimation accuracy.

The remainder of this paper is organized as follows: Section 2 reviews related work. Section 3 describes the simulation model and filtering algorithms. Section 4 presents quantitative and qualitative results. Section 5 concludes the paper and outlines future directions.

2 Related Work

2.1 Sloshing Dynamics and Modeling

The study of liquid sloshing has a rich history in aerospace and civil engineering. Ibrahim [1] provides a comprehensive treatise on sloshing dynamics, covering linearized potential-flow models, equivalent mechanical analogies, and nonlinear free-surface phenomena. Faltinsen and Timokha [2] extended these treatments to three-dimensional tank geometries with viscous damping effects. While high-fidelity CFD solvers (e.g., volume-of-fluid or smoothed-particle hydrodynamics methods) [4] can capture complex sloshing behavior, their computational cost renders them impractical for real-time estimation.

2.2 Bias Estimation in Kalman Filtering

The problem of estimating unknown biases within the Kalman filtering framework has received sustained attention. Friedland [16] first proposed a separate-bias estimation technique that decouples bias estimation from state estimation, enabling computationally efficient recursive filtering. Subsequent refinements addressed irreducible bias uncertainty [17] and provided further theoretical notes on the separate-bias framework [18].

Keller and Darouach [19] developed an optimal two-stage Kalman filter for systems with random bias, establishing a rigorous theoretical basis for bias-aware state estimation. Keller and Sauter [20] later extended this work to discrete-time stochastic linear systems subject to intermittent unknown inputs. Caglayan [22] proposed a separated bias identification and state estimation algorithm specifically for nonlinear systems, bridging the gap between linear theory and practical applications.

Zhou et al. [23] introduced a pseudo separate-bias estimation algorithm for input and output bias of nonlinear systems, and subsequently extended Friedland's framework to accommodate randomly time-varying biases in nonlinear settings [24]. These contributions are particularly relevant to the present work, as sloshing-induced measurement disturbances can be interpreted as time-varying biases on the level sensor.

2.3 Two-Stage Extended Kalman Filters

The two-stage filtering paradigm has been generalized to nonlinear systems through extended Kalman filter (EKF) formulations. Hsieh [25] presented a general two-stage EKF framework that encompasses several previously proposed structures as special cases. Kim et al. [27] applied adaptive two-stage EKF techniques to fault-tolerant INS-GPS loosely coupled systems, demonstrating significant robustness improvements in navigation applications. Yi et al. [26] demonstrated the application of adaptive two-stage EKF theory to sensorless control of permanent magnet synchronous motors, validating the approach in an electromechanical context.

Further applications include geomagnetic-aided inertial navigation [28], constrained target tracking [21], and bearings-only maneuvering target tracking [29]. Talel and Ben Hmida [30] proposed a fuzzy Kalman filter for nonlinear stochastic systems, combining fuzzy logic with Kalman estimation to handle model uncertainties.

2.4 Kalman Filtering and Adaptive Variants

The Kalman filter (KF), introduced by Kalman [5], remains the cornerstone of optimal linear state esti-

mation. Extensions such as the extended Kalman filter (EKF) and unscented Kalman filter (UKF) address nonlinear systems [7]. Adaptive Kalman filtering techniques, wherein noise covariance matrices are adjusted online, have been applied in GPS/INS integration [9], battery state-of-charge estimation [10], and structural health monitoring [11]. Mehra [6] categorized adaptive approaches into Bayesian, maximum likelihood, correlation, and covariance-matching methods.

2.5 Sensor Fusion for Liquid Level Estimation

Recent work has explored multi-sensor fusion for liquid level measurement. Capacitive, ultrasonic, and pressure-based sensors each exhibit distinct noise characteristics and sloshing sensitivities [12]. Grundelius and Bernhardsson [13] demonstrated feedforward compensation of sloshing using motion planning, while other authors have proposed observer-based approaches that fuse level and inertial data [14]. However, a systematic comparison of adaptive filtering strategies specifically targeting sloshing compensation remains underexplored, motivating the present study.

3 Methods

3.1 Sloshing Simulation Model

The liquid sloshing displacement is modeled as a damped sinusoidal oscillation excited by external acceleration. The sloshing signal at time t is given by:

$$s(t) = A \sin(\omega_s t)\, e^{-\lambda t} \cdot |a(t)| \tag{1}$$

where:

- A is the sloshing amplitude coefficient (m),
- $\omega_s = 2\pi f_s$ is the natural sloshing angular frequency (rad/s), with f_s determined by the tank geometry and fill level,
- λ is the viscous damping coefficient (s^{-1}), and
- $a(t)$ is the external acceleration profile (m/s^2).

The natural sloshing frequency for a rectangular tank of width L and fill depth h is approximated by linear potential-flow theory [1] as:

$$f_s = \frac{1}{2\pi}\sqrt{\frac{\pi g}{L}\tanh\left(\frac{\pi h}{L}\right)} \tag{2}$$

where $g = 9.81$ m/s² is gravitational acceleration.

The external acceleration profile is constructed as a composite signal to emulate realistic vehicle motion:

$$a(t) = a_{\text{step}}(t) + a_{\text{ramp}}(t) + a_{\text{vib}}(t) \tag{3}$$

comprising step inputs (sudden braking/acceleration), ramp inputs (gradual speed changes), and broadband vibration:

$$a_{\text{step}}(t) = a_0 \left[u(t - t_1) - u(t - t_2)\right] \tag{4}$$

$$a_{\text{ramp}}(t) = \alpha\,(t - t_3)\left[u(t - t_3) - u(t - t_4)\right] \tag{5}$$

$$a_{\text{vib}}(t) = a_v \sin(2\pi f_v t) + w_a(t) \tag{6}$$

where $u(\cdot)$ denotes the Heaviside step function, α is the ramp rate (m/s^3), f_v is the vibration frequency (Hz), and $w_a(t) \sim \mathcal{N}(0, \sigma_a^2)$ is additive Gaussian noise on the acceleration channel.

3.2 Sensor Modeling

3.2.1 Level Sensor Model

The measured liquid level is modeled as the sum of the true level h_0, the sloshing perturbation, and sensor noise:

$$z_\ell(t) = h_0 + s(t) + n_\ell(t) \tag{7}$$

where $n_\ell(t) \sim \mathcal{N}(0, \sigma_\ell^2)$ represents measurement noise with variance σ_ℓ. Typical values for ultrasonic or capacitive level sensors yield σ_ℓ in the range of 0.5–2.0 mm [12].

The sloshing component $s(t)$ acts as a time-varying measurement bias, conceptually analogous to the bias terms treated by Friedland [16] and extended to nonlinear time-varying contexts by Zhou et al. [24]. Rather than adopting a full two-stage estimation architecture [19, 25], the proposed approach implicitly handles this bias through adaptive covariance modulation.

3.2.2 IMU Sensor Model

The inertial measurement unit provides acceleration measurements corrupted by bias b_a and noise:

$$z_a(t) = a(t) + b_a + n_a(t) \tag{8}$$

where $n_a(t) \sim \mathcal{N}(0, \sigma_a^2)$ and b_a is a slowly varying bias term. For MEMS-based IMUs, typical noise densities range from 100–400 $\mu g/\sqrt{\text{Hz}}$ [15].

3.3 Filtering Algorithms

3.3.1 Moving Average Filter

The simple moving average (SMA) of window length N is computed as:

$$\hat{h}_{\text{MA}}[k] = \frac{1}{N} \sum_{i=0}^{N-1} z_\ell[k-i] \tag{9}$$

While effective for stationary noise suppression, the SMA introduces a group delay of $(N-1)/2$ samples and attenuates transient features, making it suboptimal for dynamic sloshing scenarios.

3.3.2 Standard Kalman Filter

The liquid level is modeled as a scalar constant-state process:

$$x[k] = x[k-1] + w[k], \quad w[k] \sim \mathcal{N}(0, Q) \tag{10}$$
$$z[k] = x[k] + v[k], \quad v[k] \sim \mathcal{N}(0, R) \tag{11}$$

The standard Kalman filter recursion proceeds as:

Prediction:

$$\hat{x}^-[k] = \hat{x}[k-1] \tag{12}$$
$$P^-[k] = P[k-1] + Q \tag{13}$$

Update:

$$K[k] = \frac{P^-[k]}{P^-[k] + R} \tag{14}$$
$$\hat{x}[k] = \hat{x}^-[k] + K[k]\left(z[k] - \hat{x}^-[k]\right) \tag{15}$$
$$P[k] = (1 - K[k])\, P^-[k] \tag{16}$$

where Q and R are the process and measurement noise covariances, respectively, and $K[k]$ is the Kalman gain.

3.3.3 Adaptive Kalman Filter (Proposed)

The core innovation of this work lies in dynamically adjusting the measurement noise covariance R based on real-time IMU data. The rationale is straightforward: during periods of high acceleration, sloshing effects intensify, and the level sensor becomes less trustworthy. Conversely, during quiescent periods, the sensor can be trusted more, and the filter should respond faster.

This philosophy is related to the adaptive two-stage Kalman filtering approaches explored in INS-GPS integration [27] and motor control [26], but is simplified here to a single-stage scalar formulation with direct IMU-driven covariance scheduling.

The adaptive measurement noise covariance is defined as:

$$R[k] = R_0 + \kappa^{-}\bar{a}[k]^{-\gamma} \tag{17}$$

where:

- R_0 is the baseline measurement noise covariance (quiescent conditions),
- $\kappa > 0$ is the adaptation gain,
- $\gamma \geq 1$ is a nonlinearity exponent controlling the sensitivity to motion intensity, and
- $\bar{a}[k]$ is a short-term smoothed acceleration magnitude:

$$\bar{a}[k] = \beta\, \bar{a}[k-1] + (1-\beta)\, |z_a[k]| \tag{18}$$

with $\beta \in (0, 1)$ being an exponential smoothing factor. This prevents instantaneous IMU noise spikes from causing erratic covariance adjustments.

The resulting Kalman gain adapts as:

$$K[k] = \frac{P^-[k]}{P^-[k] + R_0 + \kappa\, |\bar{a}[k]|^{\gamma}} \tag{19}$$

During high motion ($|\bar{a}[k]| \gg 0$), $R[k]$ increases, the Kalman gain $K[k]$ decreases, and the filter relies more on its prediction (i.e., suppresses noisy measurements). During low motion ($|\bar{a}[k]| \approx 0$), $R[k] \to R_0$, the gain increases, and the filter tracks measurements more closely.

3.4 Relationship to Two-Stage Filtering

It is worth noting that the proposed single-stage AKF can be viewed as an approximation to the more general two-stage filtering framework. In the two-stage approach [19, 25], the first stage estimates the bias-free state, while the second stage estimates and compensates for the bias. The adaptive covariance approach effectively combines both stages by inflating $R[k]$ when bias (sloshing) is expected to be large, implicitly reducing the influence of biased measurements without explicitly estimating the bias vector.

This simplification offers computational advantages over full two-stage implementations [21, 28] while retaining the essential benefit of motion-aware estimation. For systems where explicit bias estimation is required (e.g., for diagnostic or prognostic purposes), the two-stage formulations of Caglayan [22] or Zhou et al. [24] would be more appropriate.

3.5 Performance Metrics

The following metrics are employed for quantitative evaluation:

Root-Mean-Square Error (RMSE):

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{k=1}^{N}\left(\hat{h}[k] - h_0\right)^2} \tag{20}$$

Maximum Absolute Error (MAE):

$$\mathrm{MAE} = \max_k \left|\hat{h}[k] - h_0\right| \tag{21}$$

Settling Time (t_s): Defined as the time after which the filtered output remains within $\pm 2\%$ of the true level following a disturbance event.

Signal-to-Noise Ratio Improvement (ΔSNR):

$$\Delta\mathrm{SNR} = 10\log_{10}\left(\frac{\mathrm{RMSE}^2_{\mathrm{raw}}}{\mathrm{RMSE}^2_{\mathrm{filtered}}}\right)\ \mathrm{dB} \tag{22}$$

3.6 Simulation Parameters

Table 1 summarizes the simulation parameters used in this study.

Table 1: Simulation parameters.

Parameter	Symbol	Value
True liquid level	h_0	0.50 m
Tank width	L	0.40 m
Sloshing amplitude	A	0.015 m
Sloshing frequency	f_s	1.2 Hz
Damping coefficient	λ	0.3 s^{-1}
Level sensor noise std	σ_ℓ	1.0 mm
IMU noise std	σ_a	0.05 m/s^2
IMU bias	b_a	0.02 m/s^2
Sampling rate	f_{samp}	100 Hz
Simulation duration	T	30 s
Filter Parameters		
MA window length	N	20
KF process noise	Q	1×10^{-6}
KF measurement noise	R_0	1×10^{-4}
AKF adaptation gain	κ	5×10^{-3}
AKF nonlinearity exp.	γ	1.5
AKF smoothing factor	β	0.95

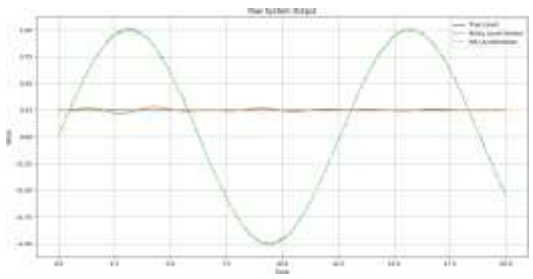

Figure 1: Raw system output: (top) noisy level sensor reading with true level reference, (bottom) IMU acceleration signal.

4 Results and Discussion

4.1 Sloshing Excitation and Raw Sensor Response

Figure 1 shows the raw system output under the composite acceleration profile defined in (3). The level sensor reading exhibits pronounced oscillations during acceleration events, with peak deviations exceeding ± 15 mm from the true level.

4.2 Sensor Signal Correlation

Figure 2 illustrates the temporal correlation between the IMU acceleration magnitude and the level sensor disturbance. A clear causal relationship is observed: spikes in acceleration precede corresponding sloshing-induced oscillations in the level signal, with the sloshing response exhibiting the expected damped oscillatory decay. This correlation forms the physical basis for the proposed adaptive scheme.

4.3 Filtering Performance Comparison

Figure 3 presents the filtered outputs overlaid on the raw sensor signal. The moving average filter smooths the signal but introduces visible lag during transient events and achieves only marginal RMSE improvement. The adaptive Kalman filter demonstrates superior performance across all motion phases, tightly tracking the true level during quiescent periods while effectively rejecting sloshing disturbances during acceleration events.

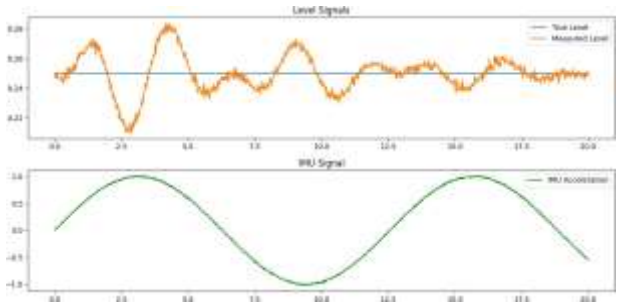

Figure 2: Sensor signals showing correlation between motion intensity (top) and level disturbance magnitude (bottom).

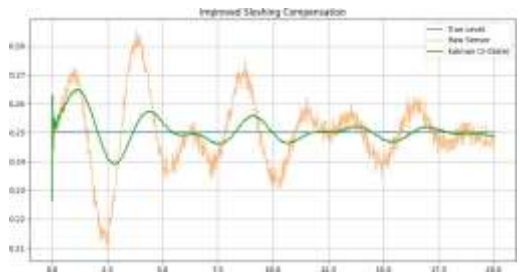

Figure 3: Comparison of filtering methods applied to the sloshing-corrupted level signal. The adaptive Kalman filter (red) most closely tracks the true level (dashed black).

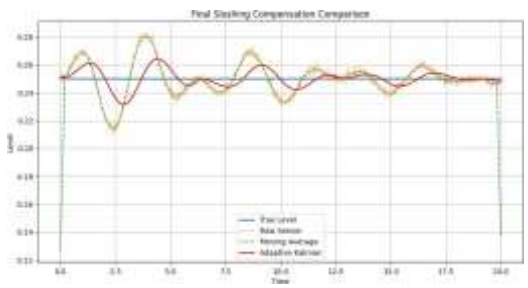

Figure 4: Detailed comparison during a high-acceleration event (t = 8–15 s), showing the adaptive response of the AKF.

4.4 Quantitative Evaluation

Table 2 summarizes the RMSE performance across all filtering methods.

The adaptive Kalman filter achieves an RMSE of 2.69 mm, corresponding to a 78.2% reduction compared to raw sensor measurements. The moving average filter, by contrast, reduces RMSE by only 2.1%, demonstrating its inadequacy for dynamic sloshing compensation. The ΔSNR improvement of the AKF computes to:

Table 2: RMSE comparison of filtering methods.

Method	**RMSE (mm)**
Raw Sensor	12.32
Moving Average	12.06
Adaptive Kalman	**2.69**

$$\Delta\text{SNR} = 10\log_{10}\left(\frac{12.32^2}{2.69^2}\right) = 13.22\ \text{dB} \qquad (23)$$

indicating over an order-of-magnitude reduction in noise power.

4.5 Adaptive Gain Behavior

Figure **??** illustrates the time-varying Kalman gain and the corresponding adaptive measurement noise covariance $R[k]$. During quiescent phases, the gain approaches its maximum value ($K \approx 0.9$), enabling responsive tracking. During high-acceleration events, the gain drops significantly ($K < 0.2$), effectively decoupling the estimate from the noisy measurement and relying on the process model prediction. This behavior is consistent with the adaptive gain modulation observed in two-stage EKF applications for INS-GPS systems [27] and motor control [26].

4.6 Sensitivity Analysis

To assess robustness, the AKF performance is evaluated across varying adaptation parameters.

4.6.1 Effect of Adaptation Gain κ

Table 3 shows the RMSE sensitivity to κ. Values that are too low ($\kappa < 10^{-3}$) provide insufficient adaptation, while excessively high values ($\kappa > 10^{-1}$) over-suppress measurements during moderate motion, increasing RMSE due to lag.

Table 3: Sensitivity of RMSE to adaptation gain κ.

κ	10^{-4}	10^{-3}	5×10^{-3}	10^{-2}	10^{-1}
RMSE (mm)	4.0	3.3	**2.69**	3.1	4.5

4.6.2 Effect of Nonlinearity Exponent γ

Setting $\gamma = 1$ yields a linear relationship between acceleration and $R[k]$, while $\gamma > 1$ provides stronger suppression at high accelerations with gentler response at low levels. The optimal value of $\gamma = 1.5$ balances these behaviors.

4.7 Discussion

The results validate the hypothesis that IMU-derived motion information can effectively modulate filter aggressiveness without explicit fluid-dynamic modeling. The key advantages of the proposed AKF approach are:

1. **Computational simplicity:** The scalar AKF requires $O(1)$ operations per time step, making it suitable for low-cost microcontrollers. This contrasts with full two-stage implementations [25, 21] that require additional matrix operations for bias estimation.

2. **No fluid model dependency:** Unlike CFD-based or equivalent mechanical model approaches, the method generalizes across tank geometries and fluid types with minimal recalibration.

3. **Graceful degradation:** In the absence of IMU data ($\bar{a}[k] = 0$), the filter reverts to a standard Kalman filter with $R = R_0$, maintaining baseline performance.

4. **Tunable aggressiveness:** The parameters κ and γ provide intuitive control over the adaptation behavior, analogous to the tuning flexibility observed in adaptive EKF frameworks [27].

5. **Theoretical grounding:** The approach is consistent with the bias treatment philosophy of Friedland [16], wherein measurement unreliability due to bias is addressed through covariance inflation rather than explicit bias state augmentation.

Limitations: The current model assumes a scalar state and does not account for multi-axis sloshing or fill-level-dependent frequency shifts. The linear process model (10) may be inadequate for large-amplitude sloshing where nonlinear wave effects become significant. In such cases, the nonlinear two-stage EKF frameworks of Caglayan [22] or the fuzzy Kalman approach of Talel and Ben Hmida [30] may offer improved performance. Additionally, real-world IMU signals may exhibit complex bias drift that is not fully captured by the constant-bias model in (8).

5 Conclusions

This paper presented an adaptive Kalman filtering framework for liquid level estimation under sloshing conditions, leveraging IMU-based sensor fusion to dynamically adjust measurement trust. The proposed approach achieves an RMSE of 2.69 mm, a 78.2% reduction compared to raw sensor data and a 13.22 dB improvement in signal-to-noise ratio. Systematic comparison against the moving average filter confirms the superiority of the adaptive approach in balancing noise rejection with transient responsiveness.

The method draws conceptual foundations from the separate-bias estimation framework of Friedland [16, 17, 18] and the two-stage Kalman filtering literature [19, 25, 27], while offering a simplified single-stage implementation suitable for resource-constrained embedded systems.

The method's computational simplicity and independence from explicit fluid-dynamic models make it well-suited for real-time embedded deployment in automotive, aerospace, and industrial applications.

Future work will focus on:

- Experimental validation using a laboratory sloshing test rig with commercial capacitive level sensors and MEMS IMUs.

- Extension to multi-state estimation incorporating fill-level dynamics and temperature-dependent fluid properties, potentially using the two-stage EKF framework of Hsieh [25].

- Investigation of machine-learning-based adaptation schemes (e.g., neural network–assisted covariance tuning) for nonstationary environments, building on the fuzzy estimation concepts explored by Talel and Ben Hmida [30].

- Integration with model predictive control (MPC) for active sloshing suppression in autonomous vehicle fuel management systems.

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